

Sundar Gurumurthy

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Education

- M.Res. Cranfield University**, Manufacturing Jan 2023 – May 2024
- **Thesis:** Understanding and Improving the Inherent Strain Method for Mechanical Analysis of Wire Arc Additive Manufacturing.
 - **Key Topics:** Residual stress prediction, distortion modeling, process-structure-property relations.
 - **Supervisors:** Dr. Yongle Sun & Dr. Pradeeptta Taraphdar
 - **Funding:** NEWAM (EPSRC Programme Grant)
- BE Birla Institute of Technology and Science Pilani**, Mechanical Engineering Aug 2017 – July 2021
- **Grade:** 7.71 / 10
 - **Class:** First Class (Equivalent to UK 2:1)

Experience

- Cranfield University**, Research Assistant Cranfield, UK
June 2024 – present
1 year 8 months
- Built Python/C++ simulation automation pipelines for thermal-mechanical modelling and process-window analysis.
 - Developed phase-transformation strain models using test-driven statistical fitting and thermal-expansion datasets.
 - Integrated laser-scan data with simulation workflows for geometric conformity assessment and deviation-driven modelling.
 - Implemented rapid reduced-order approaches producing industry-usable simulation results in minutes.
 - Developed FORTRAN heat-source models and C++ Abaqus subroutines for nonstandard material and process cases.
 - Delivered technical reports, experimental analyses, and multi-stakeholder design reviews.
- Sona Comstar**, Graduate Engineering Trainee Gurugram, IN
July 2021 – July 2022
1 year 1 month
- Performed fatigue, structural and tolerance assessments for drivetrain and differential systems.
 - Conducted forging DfM and manufacturing reviews for machining (soft / hard machining), broaching, grinding, polishing, and casting.
 - Defined operational load cases from torque maps and test data.
 - Automated CAD-FEA pipelines using Python/VBA to improve turnaround time.
 - Delivered stress reports, DFMEA documentation and supplier handover packages.
- Mercedes-Benz R&D India**, Student Trainee - Crash Structures Bengaluru, IN
Feb 2021 – June 2021
5 months
- Built LS-Dyna models for tyre, rim and composite crash scenarios.
 - Worked on Lagrangian contact optimisation for fibre-reinforced rubber composites.
 - Automated ANSA meshing using Python scripts.
 - Supported test-simulation correlation for modal and impact behaviour.

Publications

- Study of Residual Stress Using Phased Array Ultrasonics in Ti-6Al-4V Wire-Arc Additively Manufactured Components** Oct 2024
- Sundar Gurumurthy
www.mdpi.com/1424-8220/24/19/6372

Development and Testing of an Antenna Deployment System for Nanosatellite Applications

Oct 2019

Sundar Gurumurthy

www.scopus.com/record/display.uri?eid=2-s2.0-85079192397

Dynamic Active Thermal Control of a LEO Nanosatellite Based on Its Mode of Operation

Aug 2019

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www.base-search.net/Record/76e571816a3d2618dc41f6e5e604bda26a57061e95e9fb6b8cac6a70f943912f

Skills

Simulation & Modelling

Data Science

Programming Languages

FEA Solvers

CAD Tools

Experimental Methods

Awards

Student Travel Award

Aug 2019

Utah State University

Volunteer

NSS BITS Pilani, Volunteer

Pilani, India

- Tutored underprivileged teenagers enrolled under India's Right to Education Act
- Assisted in fundraising and donation drives for student scholarships

Jan 2018 – Dec 2018